Personal intelligent communications

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Intelligent network technology can rapidly deliver highly featured voice services on a personal choice basis. In practice, the services are often too complex for end users to understand and control using a basic telephone. New services are being identified that encompass electronic information and messaging as well as voice. At the same time, computer vendors are adopting new techniques allowing computers to be used more easily, such as those employed in personal digital assistants (PDAs). All these developments are leading to significant changes in service concepts, how services are accessed and controlled, and the terminal features available to both fixed and mobile users. This paper describes some of these developments and examines some of the implications for personal communications service implementors and end users.

1. Introduction

Network intelligence concepts used in telephone networks are leading to a wide range of automated services that in the past could not be implemented without human interaction with the user. Personal communications services are a family of services that are becoming increasingly important today. They are designed to match the needs of individuals who are mobile. In this context, mobile means any access mode may be used as desired, not only cellular mobile. Wired access or wireless access may be used from any terminal, in any combination with the same familiar services instantly available, leading to communication any time, any place, using any mode.

These concepts are not new. They form the basis of a set of standardised services called universal personal telecommunications (UPT), a key driver for intelligent networks. In UPT work, the terms personal mobility and terminal mobility are often used. Personal mobility refers to the user's ability to access telecommunications services from any network and terminal on the basis of a unique personal identifier, and the capability of the network to deliver those services according to the user's profile. Personal mobility concepts can be applied to both wireless and wired terminals, and result in a 'personalised' telephone. Personal mobility services are a vehicle for convergence between wired and wireless access, as illustrated in Fig 1.

Terminal mobility describes the user's ability to access services while in motion, i.e. while using wireless access, such as cordless, cellular mobile or satellite.

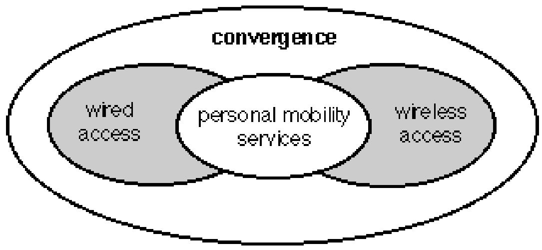

Fig 1 Wired and wireless access service convergence.

Personal mobility implies a choice. Users can have telephony services tailored to match individual needs. Thus the user has many options to choose from. Examples of this are variable incoming call routeing based on a personal number where the user defines the destination network number to be used when the personal number is dialled, or personal billing where an outgoing call can be made from any telephone and the charge billed to the user's personal account. The concepts can be further extended to allow service features to be enabled, disabled or modified as the user desires. For example, at certain times users may wish to be alerted when their bill has reached a chosen limit, while at other times this may be of no interest. The required service feature might be selected by the user and a limit entered, or deselected when not required. The information about a user's chosen combination of service features and options, and the necessary data, such as time or destination number, must be held in the network database. When changes are complete, users will expect them to have immediate effect.

These scenarios raise some interesting implementation problems, incuding those associated with user access to the database, which is a major focus of this paper. Some important questions a user might ask about their service configuration data include the following.

- How do I know what options are available?
- How do I select the options I require?
- How do I know which options are currently selected?
- How can I cope with entering the many digits required?

The simplest services may offer only a small number of feature options, where one digit is sufficient to identify the option desired, but even the simplest services may require many data digits to be entered. Table 1 illustrates the point by listing the stages that might be used to enter the data digits required to configure a user's 'profile' for a schedule-based variable-routeing service. If short codes are used to select the first and second choice destinations, in the example shown, 54 digits would be required. If short codes were not used, 68 digits would be required. Neither is acceptable when using an ordinary telephone DTMF (dual tone multi-frequency) keypad.

Table 1 Example profile entry illustrating the large number of digits required.

User entry	No of digits
Access (e.g. 0800 xxxxxx)	10
ID and PIN	20
Select option	1
Date	6
Start time	5
End time	5
First choice destination	3(10)
Second choice destination	3(10)
End	1
Total number of digits	54(68)

To select the desired option the user typically hears a voice announcement and responds by entering DTMF digits or by a spoken response. In the case of a spoken response, voice recognition in the network is required. In practice, it has been found that both suffer from being slow leading to user frustration. However, it is possible to reduce the problem by keeping the DTMF detector on line so that voice announcements can be interrupted provided the user remembers which digit to press. Even so, it is still a lengthy process if there are more than a small number of options. An acceptable number might be two or three options.

As the number of service options increases, it becomes necessary to partition them into sub-menus, just like the menu driven computer software packages widely used before graphical user interface (GUI) techniques were invented. A new problem is then found to occur when using an ordinary telephone. Voice announcements and selection digits have to be remembered, unlike when using a computer, where the menu can be viewed on the display screen. This becomes especially difficult when it becomes necessary to navigate a menu tree in order to select required options. Users often forget where they are in the tree structure, or they forget which digit they should enter next to select the option required. They are forced to traverse back up the tree again to find out what to do.

Having successfully selected the options required and entered the data digits, yet another problem occurs when users decide to make a new change. They cannot always remember which options were configured previously and which data digits were entered. It then becomes necessary to include voice announcements to read back the current profile, provide a printed copy by facsimile, or the user must write down each change as it is made.

These problems lead to user dissatisfaction. Using the service is not the pleasurable experience it should be. Instead of stimulating usage and creating revenue for the network operator the user interface problems become a barrier.

2. Users in control

Today, fixed network local exchanges support some services beyond the ability to make and receive basic telephone calls. They are specific to the access lines being used. Examples of these are:

- divert when busy,
- three-way calling,
- call charge advice.

They are invoked by entering a start digit (e.g. '*') followed by the code number for the service required and a closing digit (e.g. '#'). Analogue cellular networks offer similar services.

In the United States, customer-calling local area signalling system (CLASS) services for wired users have grown rapidly over the past decade. CLASS is defined by Bellcore as a method for transferring data between the local exchange and CLASS customer premises equipment (CPE). The services offered include calling number identification, calling name identification, automatic call-back and selective call forwarding. They use information delivered to the local exchange in the SS7 signalling channel and can be implemented using a traditional switch-based approach. They do not demand an intelligent network implementation.

Trials using screen phones have been carried out where options are selected by pushing buttons next to a list of choices shown on a display, instead of entering commands using an ordinary telephone without a display. Users find the visual interface much easier, instead of having to remember service codes and telephone numbers. Further evaluation of the results showed that an enhanced signalling protocol was required, leading Bellcore to develop a new protocol called the analogue display services interface (ADSI). Several companies are now testing ADSI telephones that allow customers to manipulate and control advanced services [1].

CLASS and ADSI have shown that more complex telephony services require a much greater level of user control than in the past, and that a simple telephone keypad without a display is not suitable. Personal mobility services incorporate a 'personalised' telephone concept, where a user perceives the same service independent of the terminal used. Screen phones can store user information such as telephone number lists, service access numbers and preferred control sequences.

If personal communications users were restricted to wired access in an intelligent network scenario, the screen phone approach offers a possible solution to the service control problem, but for the user who is not restricted to wired access it may be only a partial solution.

In the GSM system, subscriber identity module (SIM) cards are used to identify the user for billing purposes and to store personal information, user preferences and messages received via the short message service. A GSM telephone cannot be used unless a valid SIM card is inserted. This technique for achieving terminal independence could also be applied to wired telephones.

In future intelligent network implementations it can be expected that service control will be provided by communications between the service management system (SMS) in the network and service applications in the user's terminal. The service changes established via the SMS are downloaded to the service control point (SCP) for execution. Terminals are expected to employ visual representations and simple entry techniques that turn the service control problem into a pleasurable experience. Thus network intelligence can be combined with terminal intelligence to remove complexity and to make telephony service control easier.

Communications between the SMS node and the terminal application may typically take place via a modem located in the local exchange or in an intelligent peripheral (IP)/service node as illustrated in Figs 2 and 3.

In Fig 2 the network modem is co-located with the local exchange line card. A non-circuit-related signalling channel is used to carry information between the network modem and the SMS.

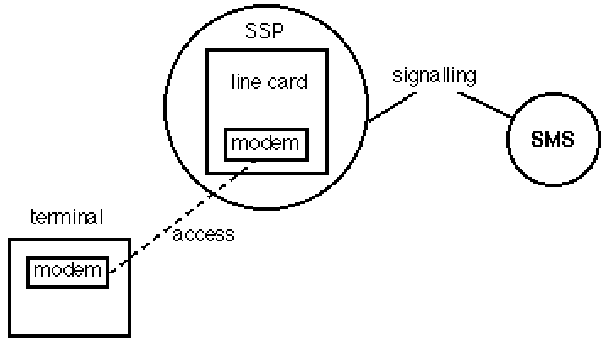

Fig 2 Service control via line card.

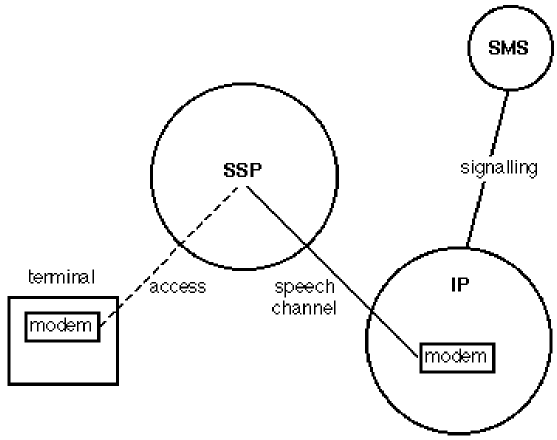

Fig 3 Service control via intelligent peripheral (IP).

Figure 3 illustrates an alternative implementation where baseband modem signals are carried in a speech channel through the switch to the IP where the network modem is located. A non circuit related signalling channel is used to carry information between the IP and the SMS.

3. Intelligent terminals

The user control problems and service options lead to a vision for the future — a single communications device or terminal that can be used for both voice and data, incorporating seamless, fully integrated communications. The term ‘personal intelligent communicator’ or ‘PIC’ is used to describe this device.

PIC characteristics include:

- seamless integrated communications access, wired or wireless,
- access to voice, electronic messaging and electronic information services,
- service feature control,
- a hand-held, pocket-sized device,
- highly integrated applications,
- a large display screen,
- a simple input device,
- long battery life.

In the PIC concept, intelligence in the terminal application software and intelligence in the network service software combine making the complexities of the technology invisible. The services become consequently easy and enjoyable to use. The concept is illustrated in Fig 4.

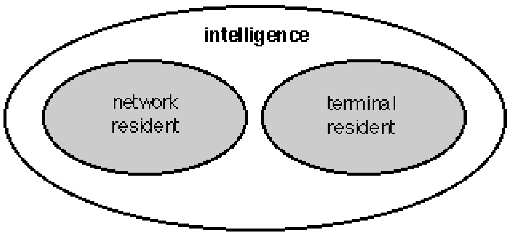

Fig 4 Combining network intelligence with terminal intelligence to remove service complexity.

In 1993 computer manufacturers launched a new kind of computer called a personal digital assistant or PDA. There are a number of products available at the time of writing this paper. They are essentially electronic personal organiser products combining computing with data communications, but do not include voice telephony. New concepts have been employed to make computing accessible to all consumers.

PDA features include:

- integrated personal organiser type applications,
- communications by facsimile and e-mail,
- pen on screen input in place of a keyboard,
- handwriting recognition,
- intuitive user interface,
- large LCD display,
- infra-red port,
- pocket size,
- battery operated,
- personal computer memory card international association (PCMCIA) card slot .

More recently, software agent concepts are leading to new approaches towards integrating terminal-resident intelligence with network-resident intelligence.

4. Personal digital assistants (PDAs)

The first generation PDA products available today are expensive. Applications may include communications, calendar, address book, note book, calculator, world clock, file manager, games, spell check and language translator. Applications have a graphical user interface designed to be intuitive and easy to use. They are described as integrated because applications can readily exchange information. Keywords can be used in one application so that information entered is automatically registered and available to other applications. For example, an appointment entered into the notepad using the keyword ‘meeting’ is automatically entered into the day planner at the time and on the day required. Simialrly, if a person’s name is entered into the day planner and their details are already stored in the address book, a simple gesture with the pen reveals the information.

Communications features can be used to create and send facsimile messages or to send and receive e-mail messages. Facsimile messages can be sent to any compatible facsimile machine, but to use e-mail an account with an e-mail service is required. The communications software allows simplified transfer of telephone numbers from other applications such as an address book, but the procedure for generating an e-mail or fascsimile and sending it often requires several steps that can be laborious. This is partly because the software is designed to communicate with what are essentially manual systems.

All PDAs and most pen computers use an unwired pen as the main input device. Alphanumeric characters are written directly on the screen. The screen incorporates a digitiser pad to detect the position of the pen and convert position to digital values that can be processed by the microprocessor. There are many technologies that might be used to detect pen position including contact, magnetic, electric, acoustic and optical. However, there are a number of constraints brought about by the need to press on the writing surface and the need to detect written characters with speed and accuracy. Of all the possible techniques, two-layer resistive, or magnetic wire grid, is most commonly used for unwired pens.

One of the unique features of PDAs is their ability to recognise handwriting. The best PDAs recognise capital letters, small letters, cursive handwriting and numbers. Others recognise only single-spaced characters and numbers, but none of the techniques in use today approach 100% accuracy. For many people accuracy is well below 100%. In many instances, training the software to recognise personal handwriting characteristics can improve recognition accuracy. Yet another approach is to recognise only the first few characters, and then select the complete word from a list of ‘best guess’ words.

For many applications full handwriting recognition is not necessary. The context of a word within an application can often be used to improve recognition accuracy. There are alternatives to handwriting recognition, such as icons, buttons, and menus or lists, where the desired option can be chosen using the pen or finger as a pointing or dragging device. All PDAs incorporate a pop-up keyboard that can be used to enter individual characters and numbers instead of using handwriting recognition. For some applications, such as diagrams and sketches, handwriting recognition can be switched off and the screen image stored as ‘ink’.

The PDA concept focuses on computing with communications, whereas the PIC concept focuses on communications with computing. At first sight this may appear to be only a subtle difference, but it is very important to recognise the shift of emphasis from computing to communications. All of the PDA features are equally desirable in the PIC, but with the addition of fully integrated communications hardware and fully integrated communications software applications in one case. Not only are the software applications integrated within the PIC, they are also tightly coupled with the communications access network and service features provided by network intelligence.

5. PCMCIA cards

The PCMCIA was set up in 1989 to establish world-wide standards for personal computer (PC) cards, and to promote the standards. PC cards are designed to meet the requirements of mobile hand-held computers for additional memory and I/O functions, such as modems and radio modules. They are small plug-in units designed to be installed or removed without having to open the computer case. The basis of the standards is a 54mm by 85.6mm card with a 68 pin connector. There are three card thickness options. Type 1 is the thinnest at 3.3mm, and is used for memory. Type 2, at 5mm is typically used to house a modem. Type 3 is the thickest at 10.5mm, and is used for memory or I/O that requires more space, such as spinning disks. Release 1 of the standard appeared in September 1990 covering memory requirements. Release 2 issued in September 1991 specified capabilities for I/O, such as modems and radio modules. There is no relationship between card type number and release number. A computer need not be limited to one PC card slot. Double or triple slots may be used in any combination of types.

6. Network services

The combinations of network-resident intelligence and terminal-resident intelligence leads to easy control of personal mobility voice services, but voice is only one of the communication modes of interest. PDAs and PICs are ideal for entering and viewing information and for creating and reading textual messages. With seamless integrated communications, mobile users can benefit from combining service control with personal mobility services and access to information, messaging and multimedia services.

Public information services today range from recorded voice announcements to on-line computerised services. Voice announcements are easy to use, but time consuming, expensive and offer minimal user control. Many telcos have trialled interactive voice services using voice recognition to enhance user control. Voice is a slow medium and should be used appropriately. Computerised services are difficult to access for those who are not computer literate, and are costly to use. There are many information services that would be desirable, if only they could be accessed and controlled easily, and the terminals and services were priced for the consumer market. Examples include electronic books, newspapers, travel information and interactive books and games. In Europe today only business and special interest groups use computerised services.

Like information services, many consumers do not use public e-mail services, probably for similar reasons of high cost, accessibility and complexity.

Sending and receiving facsimiles electronically is largely an adaptation of a manual process. The widespread use seen today results from a high penetration of paper facsimile machines in businesses and a growing number of domestic machines and computers with facsimile capability. The limitations of point-to-point facsimile are well known.

Inclusion of a facsimile capability in PDAs is perhaps a reflection of the delay that has come about in provision of low-cost but potentially superior, electronic mail services.

Paging has been very successful for many years as a very low-cost short-message service. Separate pagers have often been used to complement voice communications, particularly for mobile users. However, the latest digital cellular technologies, such as GSM, have superior short-message service capabilities that remove the need for a separate pager.

The rate that information can be delivered to and from a PDA or PIC is dependent upon terminal capabilities, network service capabilities and access channel bandwidth. Both analogue systems and end-to-end digital systems have bandwidth limitations. Today, there is sufficient bandwidth for some services, such as travel information, but greater bandwidth and more local storage in the terminal will be required before more highly featured services, such as electronic book, newspaper and interactive services, can be implemented most effectively. The higher bandwidths required for multimedia services are expected to be realised when third generation mobile technologies are implemented, and as variable bandwidth transmission systems such as ATM become established.

7. Intelligent networks

Intelligent networks can deliver complex personal mobility services with many features. Each feature may have a number of user-controllable options. Each option may require a large number of data digits to be entered.

Some control or data entry is possible using DTMF or spoken responses to voice prompts, but trials have shown that this technique is suitable for simple services only. Wired display phones using modem signalling can be used to improve the user interface but they are not mobile.

Comparison of these solutions with a PDA or PIC clearly shows that PIC offers the optimum solution for mobile users because the key focus is communications, including voice. In view of the PDA features available today, there is an obvious growing disparity between the capabilities of intelligent terminals and the services offered by intelligent networks. Not only can the terminals be used for personal mobility service control, they are ideal for access to, and control of, messaging and information services. Intelligent networks have yet to exploit these capabilities. Terminals need not be restricted to one mode of access such as wired or cellular mobile. Users can choose whichever is most convenient or economic.

Intelligent network architectures of today have the potential to do much more than provide voice telephony services. They can enable seamless convergence of communications as perceived by users by integrating service control, personal mobility services, information services and messaging services, as illustrated in Fig 5. The needs of single-mode access (wired or wireless) users and multimode access (wired and wireless) users must be fully addressed.

An appropriate distribution of terminal resident intelligence and network-resident intelligence is required. Conversion of existing networks to IN architectures, installation of new networks, or installation of overlays, is costly. It is important that the revenue-earning potential is fully exploited.

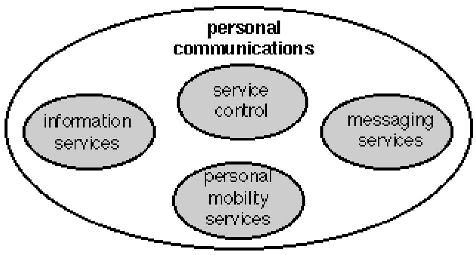

Fig 5 The personal intelligent communications vision.

8. Conclusions

Currently network intelligence is being used to implement increasingly more complex voice services in wired and cellular networks. Services are rapidly becoming too complex for users. New terminals, such as PDAs and PICs, incorporating intelligence built into soft-ware applications, can communicate with network-resident intelligence automatically, thus simplifying the user interface to the service. This combination of intelligence can greatly reduce the complexity of voice services as perceived by users, particularly when controlling or managing their personal communications. At the same time electronic messaging and information services can be readily accessed.

Personal mobility services can form a bridge between different modes of access as illustrated by convergence of wired and wireless access services.

Acknowledgement

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References

1 Caruso R E and Gary J: 'Handler', Bellcore Exchange (December 1993)

David G Smith joined BT Laboratories in 1980 working on the Monarch PABX development and then the recorded information distribution equipment (RIDE). After a further period on Monarch work, he moved to the mobile and radio networks division at BTL to lead a team developing personal mobility service concepts and demonstrators.

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